test input, full runs — small-4way

L1D 16 sets x 4, L2 64 sets x 8 · metric schema 9 · gem5 TimingSimpleCPU · 4 November 2026

35 of 35 runs completed, 0 running, 0 pending; **0 remaining**.

Program (does not depend on the cache)

run	configs done	spill/ROI	spill ld/load	spill st/store	(spill+csave)/ROI	(spill+csave) ld/load	(spill+csave) st/store
cactuBSSN	1/1	21.32%	51.31%	55.47%	21.42%	51.49%	55.80%
parest	1/1	0.22%	0.56%	1.65%	0.81%	1.48%	12.55%
deepsjeng	1/1	2.78%	9.33%	15.84%	7.31%	22.43%	46.66%
leela	1/1	0.36%	1.25%	1.66%	6.11%	15.91%	44.30%
namd	1/1	3.13%	7.91%	17.27%	3.14%	7.94%	17.37%
lbm	1/1	9.67%	27.79%	35.34%	9.67%	27.79%	35.34%
nab	1/1	7.90%	26.29%	39.86%	8.63%	27.88%	47.43%
xz	1/1	0.68%	2.35%	2.75%	3.30%	9.53%	17.13%
xz_4m_l1	1/1	0.59%	2.08%	2.43%	2.95%	8.51%	15.84%
xz_4m_l2	1/1	0.54%	1.91%	2.32%	2.73%	7.91%	15.62%
xz_4m_l3e	1/1	2.51%	9.02%	25.13%	3.76%	12.59%	42.10%
xz_4m_l4	1/1	2.12%	6.97%	14.43%	4.34%	13.23%	32.30%
xz_4m_l4e	1/1	2.28%	8.15%	25.17%	3.39%	11.30%	42.07%
xz_1m_l0	1/1	0.62%	2.50%	1.68%	3.69%	10.95%	17.49%
xz_1m_l1	1/1	0.58%	2.38%	1.48%	3.48%	10.36%	16.63%
xz_1m_l2	1/1	0.57%	2.35%	1.45%	3.39%	10.13%	16.34%
xz_1m_l3e	1/1	2.59%	9.35%	16.76%	4.60%	15.17%	33.91%
xz_1m_l4	1/1	2.01%	6.65%	9.30%	5.08%	15.11%	26.95%
xz_1m_l4e	1/1	2.43%	8.77%	16.84%	4.27%	14.07%	33.92%
imagick	1/1	0.02%	0.08%	0.05%	4.16%	11.90%	27.17%
blender	1/1	1.16%	2.72%	4.44%	9.37%	20.20%	39.50%
povray	1/1	0.27%	0.45%	0.93%	14.17%	22.21%	55.05%
omnetpp	1/1	0.09%	0.19%	0.36%	12.82%	28.51%	53.40%
xalancbmk	1/1	0.23%	0.53%	0.92%	8.96%	16.83%	45.24%
x264	1/1	6.24%	13.47%	39.69%	8.48%	18.06%	54.69%
gcc	1/1	0.84%	2.18%	3.58%	12.72%	34.38%	52.28%
mcf	1/1	1.58%	2.11%	16.47%	2.74%	3.87%	27.40%
perlbench_makerand	1/1	0.08%	0.00%	0.50%	18.63%	28.89%	59.00%
bwaves_1	1/1	3.55%	9.47%	18.87%	3.55%	9.47%	18.87%
bwaves_2	1/1	3.55%	9.47%	18.87%	3.55%	9.47%	18.87%
exchange2	1/1	5.03%	10.75%	12.58%	6.23%	13.14%	15.81%
fotonik3d	1/1	3.84%	10.81%	9.25%	3.87%	10.86%	9.40%
roms	1/1	4.79%	15.22%	9.67%	4.88%	15.37%	10.29%
wrf	1/1	0.91%	3.41%	2.49%	1.59%	4.94%	7.83%
cam4	1/1	2.11%	7.90%	7.04%	4.52%	14.20%	21.40%
pooled		5.17%	12.33%	27.44%	7.17%	16.47%	39.99%
mean		2.78%	7.88%	12.36%	6.35%	15.89%	31.03%

Miss rates

run	state	L1D	spill ld L1	other ld L1	spill st L1	other st L1	L2	spill ld L2	spill st L2
cactuBSSN	done	15.30%	4.85%	34.46%	4.27%	12.94%	65.69%	12.63%	8.16%
parest	done	10.21%	2.85%	10.95%	1.61%	2.08%	65.62%	1.84%	6.14%
deepsjeng	done	8.90%	2.01%	11.49%	7.25%	4.47%	7.54%	2.46%	14.06%
leela	done	5.85%	4.14%	6.65%	0.42%	3.69%	24.20%	0.00%	96.54%
namd	done	11.71%	2.78%	14.82%	1.14%	2.65%	28.28%	8.43%	19.85%
lbm	done	21.66%	0.00%	7.78%	0.68%	73.96%	97.36%	—	10.39%
nab	done	5.33%	0.51%	8.40%	0.31%	0.78%	6.02%	0.37%	3.38%
xz	done	5.69%	0.01%	7.48%	0.21%	2.53%	29.53%	0.14%	20.53%
xz_4m_l1	done	7.17%	0.01%	8.66%	0.22%	4.55%	39.21%	0.14%	27.77%
xz_4m_l2	done	9.13%	0.01%	11.23%	0.24%	5.05%	45.18%	0.14%	27.73%
xz_4m_l3e	done	6.35%	1.04%	7.55%	0.50%	4.39%	20.93%	0.00%	2.44%
xz_4m_l4	done	7.39%	0.98%	9.11%	0.99%	4.63%	34.26%	0.00%	7.15%
xz_4m_l4e	done	6.20%	1.15%	7.20%	0.52%	4.49%	19.17%	0.01%	2.14%
xz_1m_l0	done	5.14%	0.01%	7.30%	0.70%	1.47%	33.71%	0.46%	8.41%
xz_1m_l1	done	5.81%	0.01%	7.91%	0.67%	2.21%	39.07%	0.46%	9.95%
xz_1m_l2	done	6.07%	0.01%	8.24%	0.68%	2.30%	39.07%	0.46%	10.02%
xz_1m_l3e	done	5.46%	0.74%	7.12%	0.68%	2.66%	21.23%	0.00%	1.45%
xz_1m_l4	done	5.83%	0.70%	7.91%	1.36%	2.59%	28.43%	0.00%	2.97%
xz_1m_l4e	done	5.47%	0.83%	6.98%	0.68%	2.71%	19.76%	0.00%	1.33%
imagick	done	4.20%	8.47%	3.54%	68.44%	5.67%	85.57%	42.42%	55.53%
blender	done	5.06%	3.27%	5.21%	3.87%	4.93%	20.38%	1.08%	1.02%
povray	done	9.13%	6.99%	11.06%	19.61%	4.24%	45.58%	71.08%	69.40%
omnetpp	done	11.85%	0.05%	16.10%	0.82%	3.96%	28.01%	97.10%	3.28%
xalancbmk	done	13.28%	9.92%	16.46%	18.32%	4.61%	34.60%	4.26%	4.23%
x264	done	1.97%	0.22%	2.39%	0.89%	1.97%	31.00%	2.36%	14.53%
gcc	done	6.10%	0.33%	7.84%	1.48%	3.82%	36.94%	10.70%	17.33%
mcf	done	24.85%	4.00%	27.84%	7.21%	9.72%	77.84%	8.59%	17.95%
perlbench_makerand	done	3.76%	0.00%	5.10%	6.66%	1.01%	0.00%	—	0.00%
bwaves_1	done	40.95%	3.81%	45.81%	0.02%	43.46%	31.23%	6.87%	3.80%
bwaves_2	done	40.95%	3.81%	45.81%	0.02%	43.46%	31.23%	6.87%	3.80%
exchange2	done	2.01%	1.74%	2.56%	2.98%	1.15%	1.35%	0.17%	0.54%
fotonik3d	done	11.96%	3.82%	16.00%	0.01%	4.87%	61.79%	1.82%	21.07%
roms	done	14.94%	3.05%	18.09%	0.79%	12.61%	53.16%	11.50%	17.99%
wrf	done	11.55%	6.88%	12.11%	4.95%	10.35%	34.19%	9.04%	13.39%
cam4	done	9.07%	1.79%	9.16%	2.70%	10.78%	59.49%	10.34%	34.15%
pooled		7.58%	1.85%	9.44%	1.85%	5.88%	44.14%	9.85%	10.96%
mean		10.47%	2.31%	12.47%	4.63%	8.76%	37.05%	9.45%	15.95%

Formulas

column	formula	aciklama
spill/ROI	`((spill_load + spill_store) / roi_total_insts) * 100`	ROI komutlarinin yuzde kaci spill load veya spill store
spill ld/load	`(spill_load / roi_load) * 100`	ROI load'larinin yuzde kaci spill reload
spill st/store	`(spill_store / roi_store) * 100`	ROI store'larinin yuzde kaci spill store
(spill+csave)/ROI	`((spill_load + spill_store + csave_load + csave_store) / roi_total_insts) * 100`	ayni oran, callee-save save/restore dahil
(spill+csave) ld/load	`((spill_load + csave_load) / roi_load) * 100`	ROI load'larinin yuzde kaci spill reload veya callee-save restore
(spill+csave) st/store	`((spill_store + csave_store) / roi_store) * 100`	ROI store'larinin yuzde kaci spill store veya callee-save save

column	formula	aciklama
L1D	`(L1D misses / L1D accesses) * 100`	Tum load ve store access'lerinin (spill + other) L1D miss rate'i: L1D'nin bu konfigurasyondaki genel performansi.
spill ld L1	`(spill load L1D misses / spill load L1D accesses) * 100`	Spill load'larin yuzde kaci L1D'de miss oldu. Genelde dusuktur: spill slot'u kisa sure once yazildigi icin cache line cogu zaman hala L1D'dedir.
other ld L1	`((load misses - spill load misses) / (load accesses - spill load accesses)) * 100`	Other load'larin (spill olmayan load'lar) L1D miss rate'i. spill_ld_L1miss ile yan yana bakilir: spill load'lar normal veriden daha mi az, daha mi cok miss oluyor?
spill st L1	`(spill store L1D misses / spill store L1D accesses) * 100`	Spill store'larin yuzde kaci L1D'de miss oldu. Cache write-allocate: miss olan bir store once cache line'i L2'den getirir, sonra yazar.
other st L1	`((store misses - spill store misses) / (store accesses - spill store accesses)) * 100`	Other store'larin (spill olmayan store'lar) L1D miss rate'i; spill_st_L1miss ile karsilastirmak icin.
L2	`(L2 misses / L2 accesses) * 100`	L2'ye gelen tum isteklerin (L1D ve L1I miss'leri) yuzde kaci L2'de de miss oldu ve DRAM'e gitti.
spill ld L2	`(spill load L2 misses / spill load L2 accesses) * 100`	L1D'de miss olan spill load'larin yuzde kaci L2'de de miss oldu. Payda sadece L1D'yi miss eden spill load'lardir, butun spill load'lar degil.
spill st L2	`(spill store L2 misses / spill store L2 accesses) * 100`	L1D'de miss olan spill store'larin yuzde kaci L2'de de miss oldu (payda yine sadece L1D miss'leri).

spill = compiler-confirmed register-allocator spill (GCC `-dP`), csave = callee-saved register save/restore (CFI), both inside the ROI only. pooled = sum of numerators / sum of denominators over the completed runs; mean = average of the per-run percentages. Miss rates are per cache access; an L2 rate's denominator is the L1D misses of that access type.